

HRADEC KRALOVE, Czech RepublicWMO: **116480**Lat: **50.250N**Lon: **15.850E**Elev: **801**StdP: **14.28**Time Zone: **1.00 (EUC)**Period: **82-93**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.6	9.1	-0.4	5.5	3.0	5.4	7.5	9.9	28.7	33.1	25.3	32.6	3.5	0	0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	84.8	66.5	81.6	65.2	78.4	63.9	68.4	81.1	66.6	77.8	65.3	75.5	7.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.2	92.7	72.2	62.7	87.9	70.9	61.3	83.5	69.3	33.1	81.9	31.6	77.7	30.5	75.7	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.8	18.4	16.3	0.5	92.6	9.1	3.6	-6.0	95.2	-11.3	97.3	-16.5	99.3	-23.1	101.8		
(5)			0.1	72.1	9.1	1.4	-6.5	73.2	-11.8	74.0	-16.9	74.8	-23.5	75.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	29.4	29.8	38.6	47.0	56.0	60.3	64.7	64.5	57.3	48.3	37.5	32.2		
	DBStd	14.63	10.58	8.92	6.69	7.01	6.36	5.84	5.53	6.09	5.31	6.97	6.47	7.54		
	HDD50	2770	638	567	356	141	22	1	0	0	3	114	376	552		
	HDD65	6652	1103	987	818	541	283	163	76	85	237	516	826	1017		
	CDD50	1763	0	0	3	50	207	311	455	451	223	63	0	0		
	CDD65	170	0	0	0	0	4	23	66	71	6	0	0	0		
	CDH74	1837	0	0	0	9	78	239	706	724	79	2	0	0		
	CDH80	446	0	0	0	0	5	38	186	210	7	0	0	0		
	WSAvg	6.4	6.6	7.0	7.4	7.0	6.7	6.1	5.8	5.1	5.7	6.2	6.5	6.3		
	PrecAvg	25.8	2.0	1.7	1.9	1.3	2.5	2.6	3.3	2.8	2.0	1.7	1.9	2.1		
(15) Precipitation	PrecMax	32.6	3.9	4.5	5.3	2.3	5.2	5.2	6.7	6.2	4.6	5.1	3.8	4.4		
	PrecMin	20.0	0.4	0.3	0.5	0.2	0.8	1.0	0.6	0.6	0.4	0.1	0.0	0.9		
	PrecStd	3.6	0.9	0.9	1.2	0.6	1.1	1.0	1.6	1.4	1.2	1.0	0.9	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	54.5	62.2	75.2	79.9	84.2	89.7	91.5	80.3	71.7	54.8	50.3		
		MCWB	46.9	47.5	50.6	55.2	62.4	66.6	67.8	69.5	64.3	60.0	50.0	45.5		
	2%	DB	47.2	48.2	56.8	70.4	75.9	80.3	84.5	85.9	76.1	66.1	51.5	46.7		
		MCWB	43.5	43.1	48.6	54.7	59.8	64.5	66.3	67.2	62.6	57.8	48.1	44.1		
	5%	DB	43.9	43.0	52.2	65.2	72.8	76.8	81.5	81.7	72.4	62.4	48.8	44.6		
		MCWB	41.1	39.7	45.8	52.5	58.6	63.1	65.5	65.3	61.1	54.9	45.8	42.8		
	10%	DB	41.2	40.0	49.0	60.1	69.4	72.9	77.9	78.0	68.1	59.1	46.3	41.9		
		MCWB	39.0	37.5	43.8	49.9	57.5	60.9	64.2	64.1	59.0	53.6	44.0	40.1		
	0.4%	WB	46.7	48.5	52.2	57.3	64.6	69.7	70.8	70.7	66.0	61.5	51.1	46.7		
		MCDB	51.0	52.8	58.2	69.8	75.0	79.7	83.6	85.6	76.7	69.0	53.6	49.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	43.9	43.6	49.0	55.2	62.1	66.5	68.4	68.2	63.9	58.6	48.3	44.7		
		MCDB	46.5	47.8	55.2	68.2	72.4	76.8	81.0	82.1	72.6	65.0	50.8	46.4		
	5%	WB	41.5	40.1	46.8	53.2	60.2	64.4	66.6	66.5	62.0	56.4	45.9	42.6		
		MCDB	43.8	42.5	51.6	64.2	69.7	74.1	77.8	77.9	70.1	61.3	48.4	44.5		
	10%	WB	39.0	37.7	44.3	50.8	58.4	62.2	65.1	65.2	59.9	54.1	44.1	40.1		
		MCDB	41.1	39.8	48.2	58.8	66.8	70.3	75.3	75.4	67.2	58.2	46.1	41.9		
	5% DB	MDBR	9.7	11.3	13.4	17.3	19.0	17.4	19.5	19.6	17.2	15.6	9.5	7.6		
		MCDBR	11.0	14.5	21.1	26.1	26.1	25.4	27.3	27.9	24.6	20.6	13.2	9.4		
		MCWBR	8.9	10.7	13.2	11.9	12.1	11.2	11.2	11.1	11.7	12.3	9.9	7.7		
		MCWBR	10.9	14.1	19.1	24.2	22.7	21.9	24.0	23.7	21.9	19.1	12.0	9.3		
(35) Mean Daily Temperature Range	5% WB	MCWBR	9.0	11.0	12.6	11.6	11.2	10.8	10.5	10.4	11.3	12.2	9.4	7.8		
	Clear-Sky Solar Irradiance	taub	0.304	0.328	0.369	0.414	0.416	0.415	0.432	0.425	0.389	0.363	0.337	0.301		
		taud	2.337	2.308	2.259	2.197	2.235	2.268	2.227	2.263	2.334	2.375	2.394	2.380		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	230	254	262	262	267	268	261	256	253	238	216	214		
		Edh at Noon	21	28	35	42	42	41	42	39	32	26	20	18		
(44) All-Sky Solar Radiation	RadStd		265	509	832	1309	1580	1689	1599	1424	1012	585	280	196		
			46	75	132	169	183	146	182	137	141	86	35	20		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (51 neighbors)	+0.85	N/A	N/A	N/A	+0.65	+0.61	N/A	N/A	+110	+41	

Nomenclature: See separate page